作业 4

Design a uniform linear scanning array whose maximum of the array factor is 60^0 from the axis of the array ($\theta_0 = 60^0$). The desired half-power beamwidth is 13^0.

Determine

1. The range of *d* for no grating lobes
2. the excitation of the elements (amplitude and phase),
3. number of array elements
4. Chose d=0.5λ, us HFSS to simulate this array and calculate its Directivity (in dB), where the element can be dipole.

From the figure, $\theta_0=60^0$ & HPBW=13^0 ->$Nd=5\lambda$.

For no grating lobes, $d<2/3\lambda$.

Chose $d=\lambda/2$, so $\beta=-90^0$, $N=5\lambda/d=10$.

